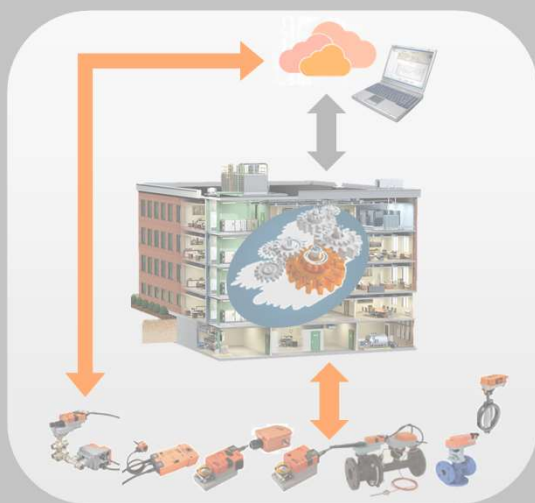


KNX - History



**High-class family homes /
Mid-sized commercial buildings**

**Originally: For Electrical Installer
HVAC with KNX 2019: Rising**

**Integrated Application + Controls
Very stable protocol**

Many KNX Manufacturer

Why do we need KNX...?

He has a **real problem**....



KNX - a **Standard** for Building Technology

Originally by Electro domain on site – now also with HVAC



He`s got a solution!



KNX - a **Standard** for Building Technology



450 Manufacturers
8000 Certified Products
One Standard: ISO/IEC 14543
One Tool: ETS
All Media: TP, PL, RF, IP

Join us
www.knx.org



KNX Certified Products

One of the strengths of the KNX system, is that any product labeled with the KNX trademark is not a mere declaration of the manufacturer but is based on conformity testing carried out by KNX accredited third party test labs. During these tests, it is not only checked that the device supports the KNX protocol but that its useful data is coded according to the KNX standardized Data types.

This results in devices of different manufacturers and different applications that can be combined to a working installation. Moreover, most of these devices can be handled by the ETS Software.

Interested in the KNX Certification Process, please see the page Certification of Products

Search Engine KNX registered and certified devices

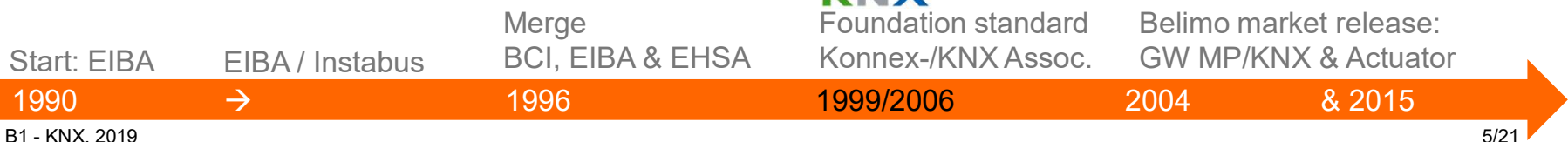
The underneath search engine allows you to search for KNX registered and certified devices according to product families and product types.

Select first the product family of interest to you (e.g. Illumination), after which you can select for the desired product type (e.g. dimmer); the search engine will now list you all KNX manufacturers with registered/certified KNX dimmers, as well as the order number of these devices.

HVAC
 Valve

Total: 48 products

Company	Product Name	Order Number
ABB Stotz-Konzepte GmbH	STK17.1 Electromotor Valve Drive	3CDS 120 004 R0011
Albrecht Jung GmbH & Co. KG	Mosor Ventiltrieb mit Regler	21775DR
Albrecht Jung GmbH & Co. KG	Stellantrieb	21775 DV
Albrecht Jung GmbH & Co. KG	aktuator	21765V
BELIMO Automation AG	...24A-KNX-Actuator	...24A-KNX
BELIMO Automation AG	...V D3-KNX-WV-Controller	...V D3-KNX
BELIMO Automation AG	EP-R-MP / P-W-E-MP, EPV	EP-R-MP / P-W-E-MP



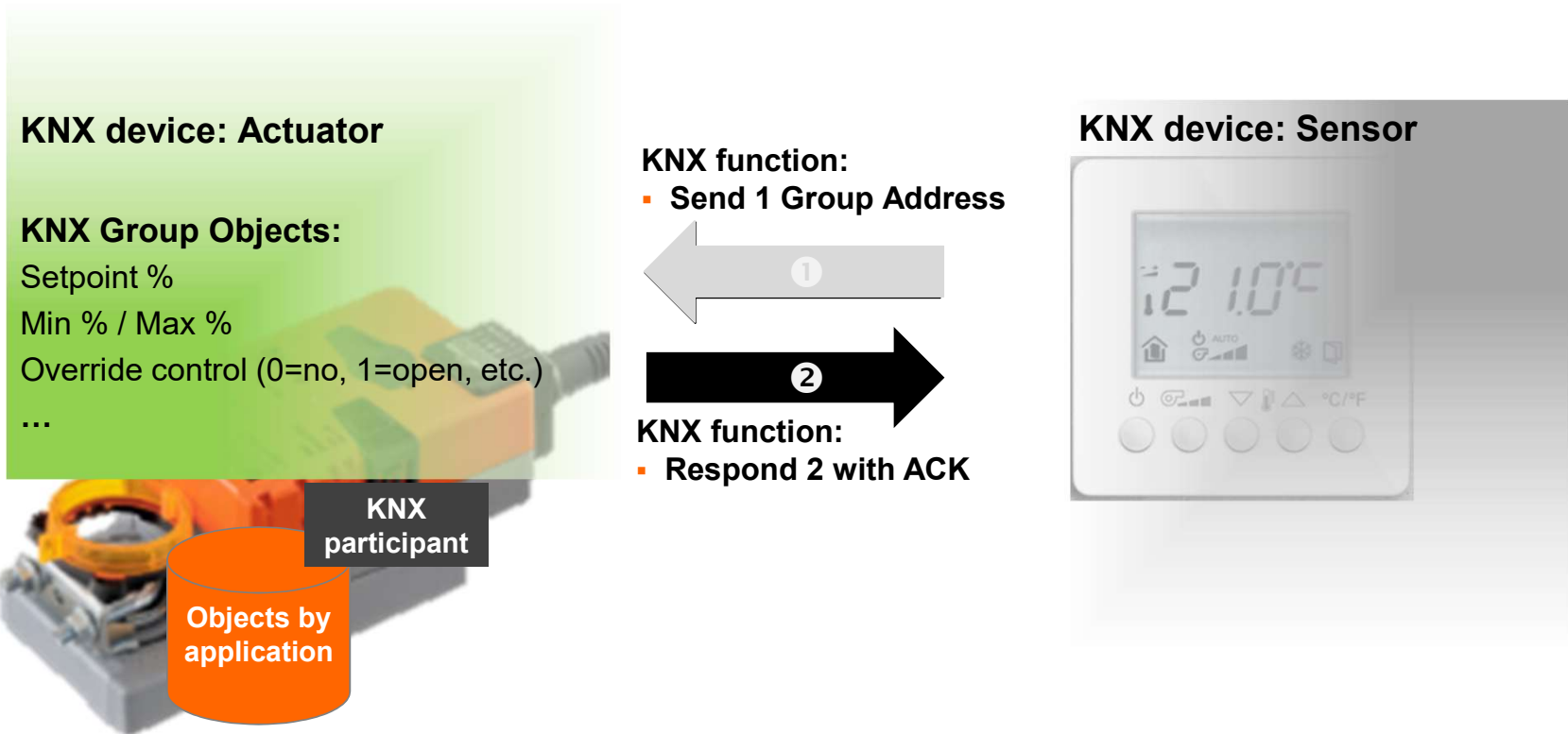
What is KNX?



Fieldbus for Light, Blind & HVAC control

- **Multi-vendor standardized** (specifications on the Internet: www.knx.org)
 - KNX part of European EN50090 standard
 - KNX part of International ISO/IEC 14543 standard
- **Multi-vendor tool (ETS) for commissioning**
- **Each device is provided with an additional application program**
 - Must be provided by the device manufacturer (e.g. Belimo)
 - Needs to be downloaded by first commissioning
 - Defines how the KNX device will work (as sensor or actuator, control logic, etc.)
 - Includes all information, that the device acts independently without any other controls

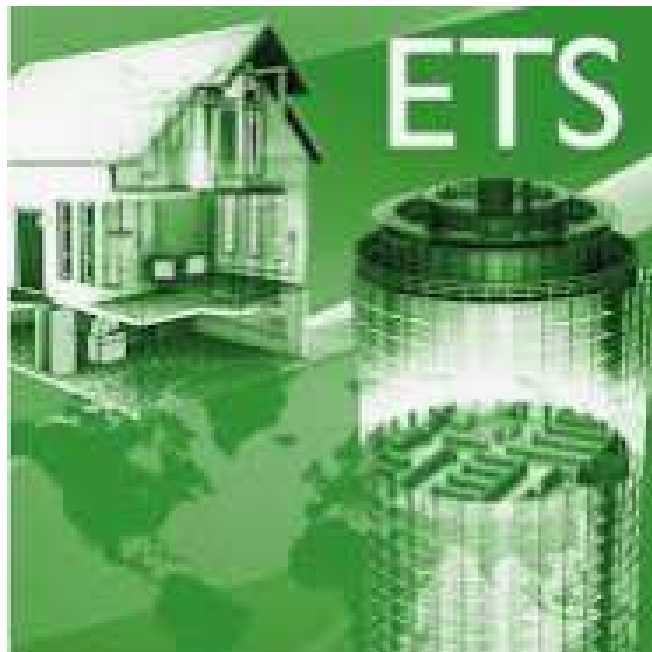
KNX: Device with Application Program



- Sensor & Actuator communication done by KNX “Group Addresses”
- Behind each “Group Address” there is a interconnected KNX “Communication Object”



Demo with Tool Software



KNX: Sensor – Actuator communication



KNX protocol

Language: Transport Options



Transport options:

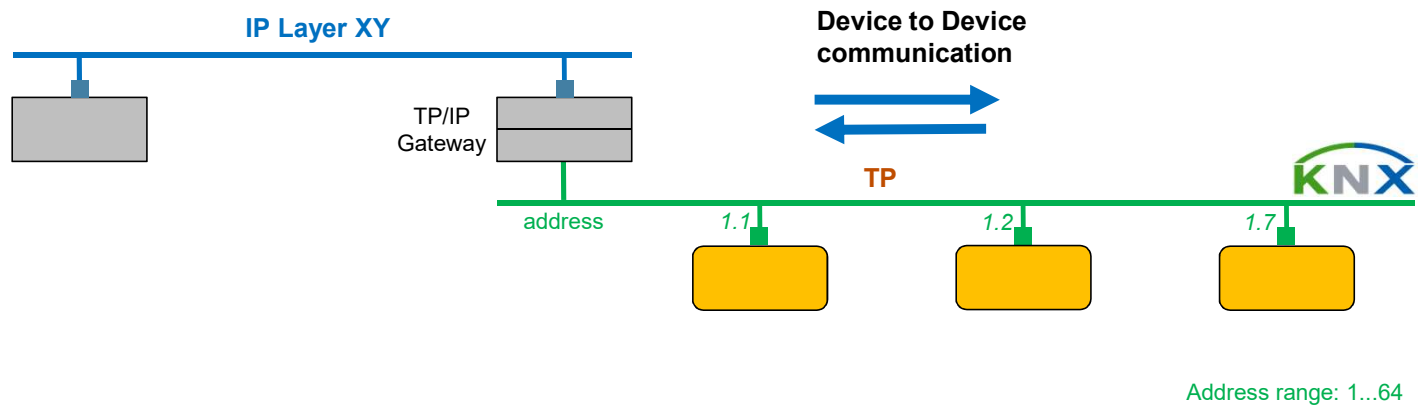
KNX TP
KNX RF
KNX IP
KNX PL



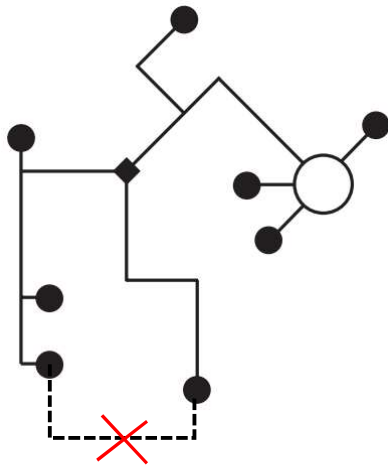
KNX Address



- 1...64 KNX address
- 15 is used for broadcasts

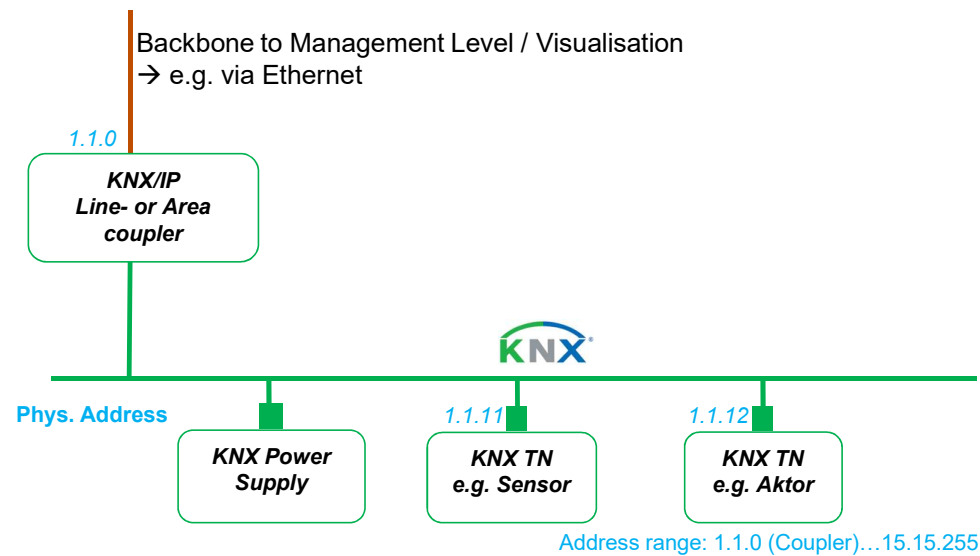


Twisted Pair Wiring

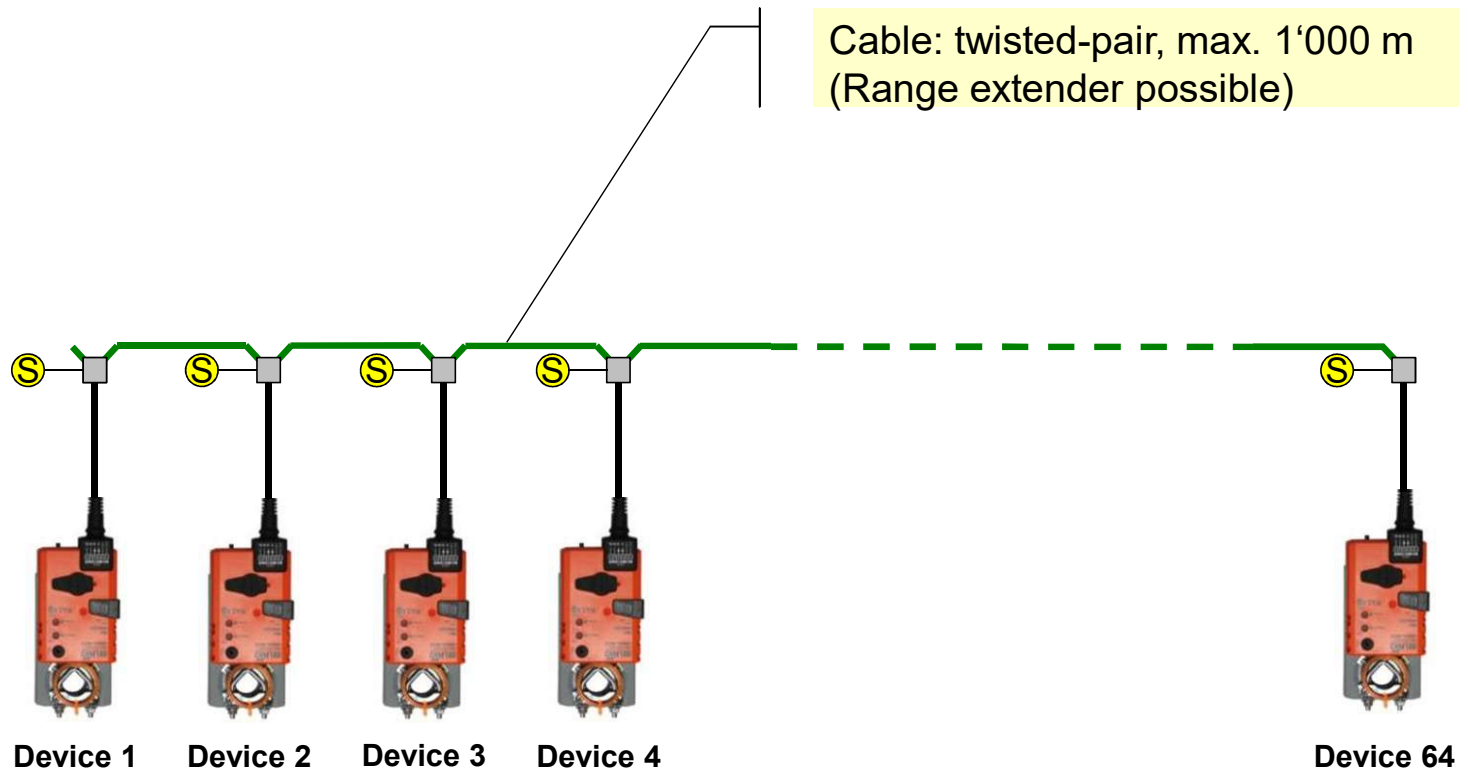


Star, ring, tree or mixed forms are permissible.

There is only one restriction with the network topology:
no closed ring are allowed!



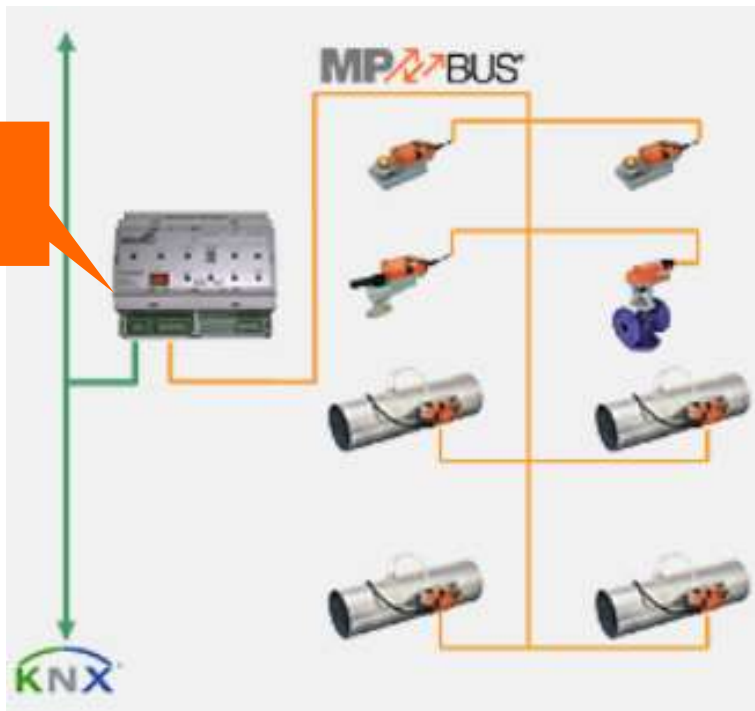
Bus line - classical topology



Power supply: KNX DC 29V as separated supply for bus communication

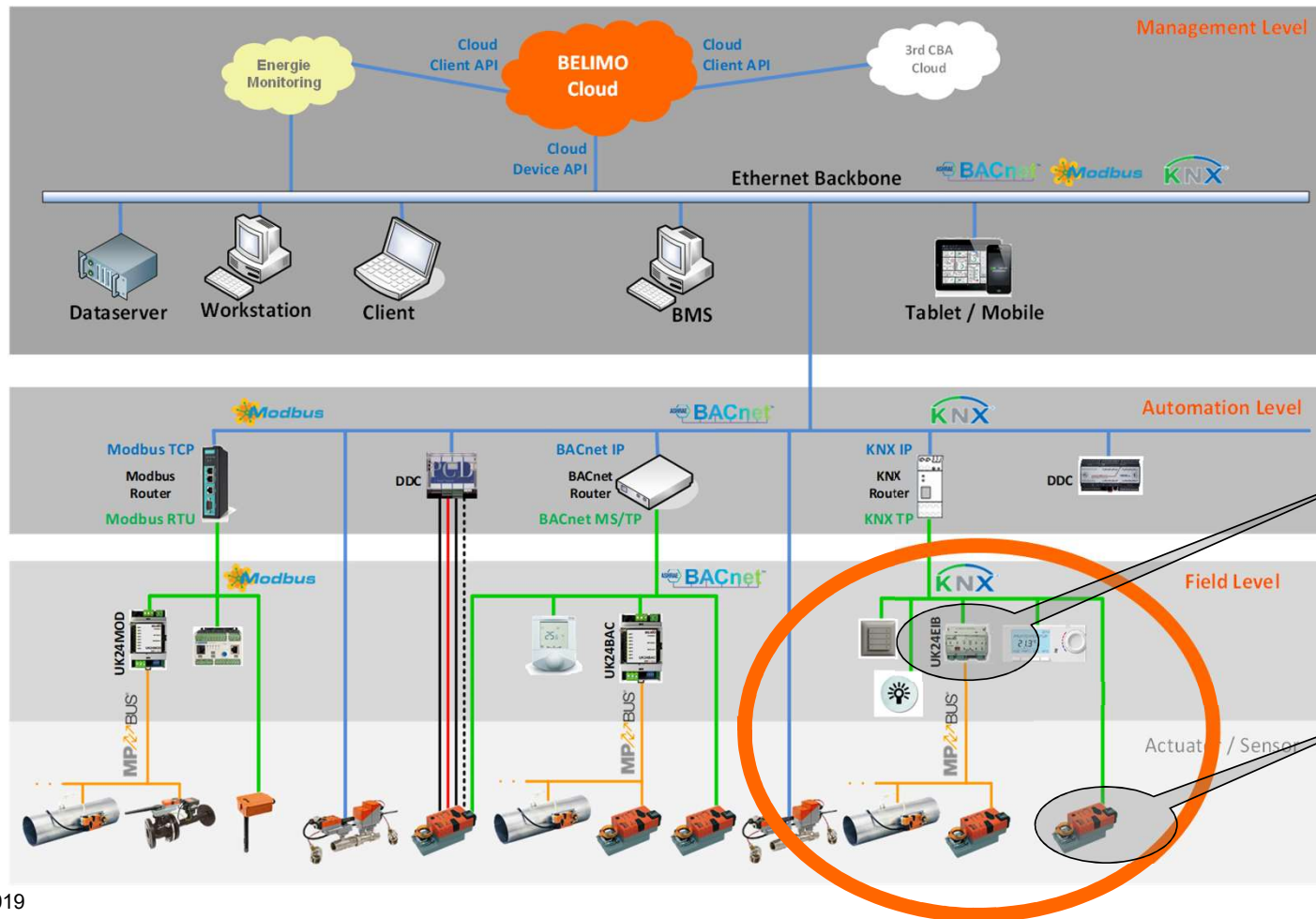
Example Gateway

**MP-KNX
Gateway**



Attention: Not all actuators / and no sensors are supported by the KNX gateways!

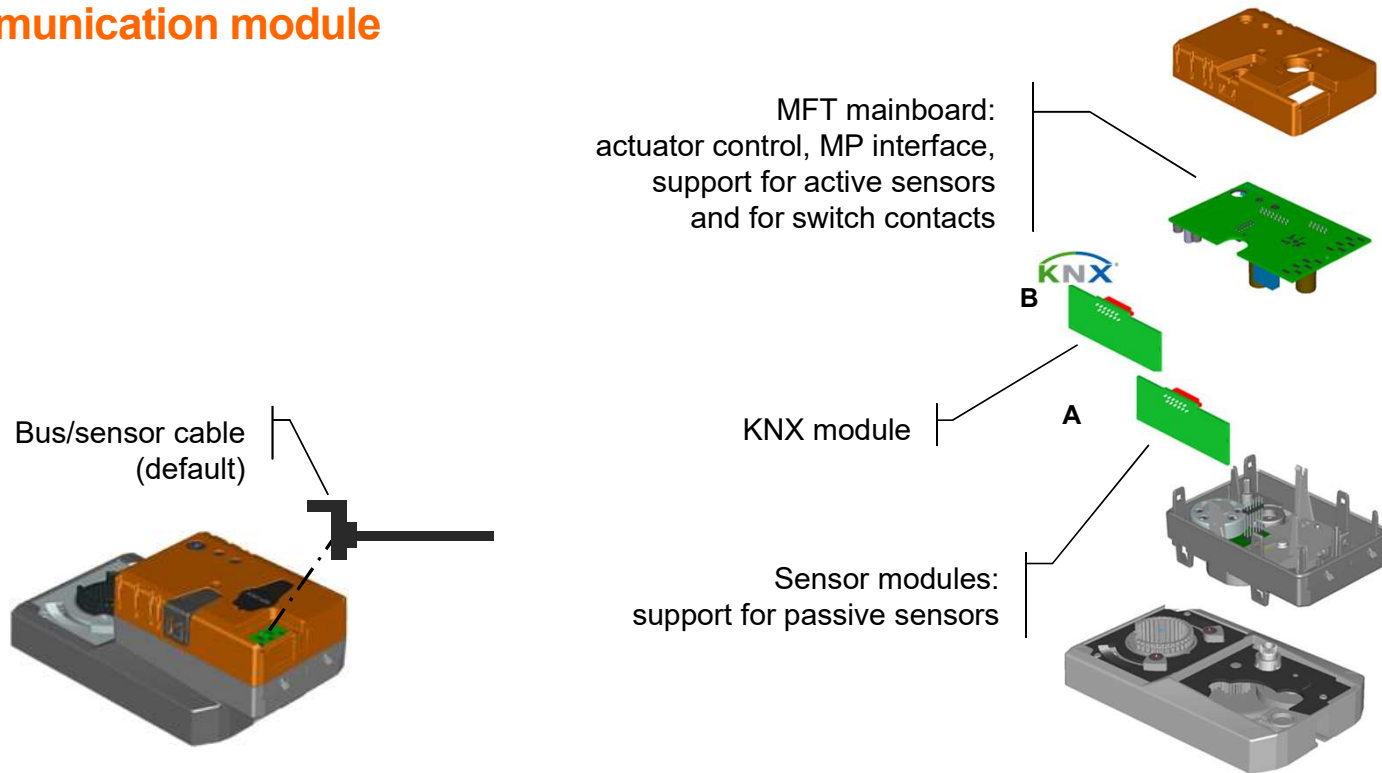
KNX @Belimo



How does KNX get on the Standard Actuators?

Belimo actuator: MFT platform example:

KNX via communication module



KNX@Belimo Product Portfolio



		
L...24A-KNX N...24A-KNX	VAV-Compact	UK24EIB



Portfolio Actuators KNX

Bus Watchdog



Goal of the Bus Watchdog:

- **Check** whether Actuator is always **controlled (Setpoint periodically set)**
- If the actuator gets no setpoint-values anymore during **a specified amount of time**, then the Bus Watchdog is activated and the actuator goes to a **predefined position**.
- Available with KNX actuators, Watchdog time configurable

Bus@Belimo Portfolio in Detail

Bus Brochure and Device Selection Tool



Products & Solutions | Documentation

- General Air Solutions
- Water Solutions
- Sensors
- Safety Solutions
- Room & System Solutions (incl. VAV and room temperature controllers)
- Bus & System Integration**
- Reference reports
- Send PDF data sheets

Download Adobe Acrobat Reader

Get Adobe Reader

Bus & System Integration

- Belimo
- LonWorks
- Modbus
- KNX
- BACnet
- Profibus
- Belimo
- MFT
- Accessories
- Phased
- Addition
- Reference

Overview

Flyer

Belimo M

- Info b
- Introd
- MP-B
- MP-Bus®: Cooperation partner (PDF - 363 kb)
- VRP-M System - MP-Bus Integration (PDF - 361 kb)
- ZFP2-MP: Product Information - Cabling solution for MP bus (PDF - 516 kb)

Bus solutions from Belimo.
The best connection.

BELIMO

View Project
mark this p
ish | Deutsch
Drehe | L
Go

Device selection tool

Filter bus devices

Direct integration

- ☐ MP-Bus
- ☐ Modbus
- ☐ BACnet
- ☐ LON
- ☒ **KNX**
- ☐ Cloud

MP-Bus with gateways

- ☐ via UK24MOD
- ☐ via UK24BAC
- ☐ via UK24LON
- ☐ via UK24EIB

Sensor connection

- ☐ Active (0-10 V)
- ☐ Active (0-32 V)
- ☐ Passive (e.g. PT1000)
- ☐ Switching contact

Application

- ☐ Air
- ☐ Safety
- ☐ VAV
- ☐ Water

Fail-safe function

- ☐ with
- ☐ without

Type

- ☒ Actuator
- ☐ Field device
- ☐ Sensor

Reset

6 suitable devices

LHV-D3-KNX	Volumetric flow controller N AC/DC 24 V, communication KNX (SIP54, Manual override w locked Connection Cable 1 m F manufacturer of your V
LM24A-KNX	Rotary actuator 5 Nm AC/DC 24 V, communication KNX (SIP54, Manual override w locked Connection Cable 1 m F
LMV-D3-KNX	Volumetric flow controller AC/DC 24 V, communication KNX (SIP54, Manual override w locked Connection Cable 1 m F Mechanical interface Un 6...20 mm
LR24A-KNX	Rotary actuator 5 Nm AC/DC 24 V, communication KNX (SIP54, Manual override w locked Connection Cable 1 m F
NMV-D3-KNX	Volumetric flow controller AC/DC 24 V, communication

KNX – Physical Addressing & Parametrizing



KNX Addressing for Belimo Devices		
Phys. Address	e.g. 1.1.10	Every KNX device requires a unique physical address from the moment it is put into operation.
KNX Seriennummer	00 0A 2C 75 0C 12	Since 2016, it has also been possible to do the commissioning using the unique KNX serial number.
Application-program	E.g. LMV-D3-KNX: BelimoActuator_DB_12.knxprod	Each manufacturer offers an application program (defines the task of the device) for each kind of KNX device, which must be loaded first into the via a KNX interface.
Parameterizing	Data Point Examples: 1bit → open/Close or up/down 1byte → 0...100% Damper position...	The communication objects in a KNX device are adapted once: communication properties are e.g. "may send / may receive / must be mute / etc."
Baud rate	9 ' 600	9600 Baud (Twisted Pair) 2400 Baud (Power Line) 16385 Baud (Radio)



Optional: Use Belimo PC-/OEM Tool for Belimo intern Parameters



Start ETS Addressing Appl.-Progr. Parameterizing Function test



1. Config. of KNX Interface (e.g. KNX USB)
2. Connect to KNX Bus
3. Commissioning Devices (push key «Addressing»)



Any questions?

